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MEC 424 - PIPING DESIGN NOTES

Hazen-Williams Equation

The Hazen-Williams formula is an empirically derived equation for circular conduits flowing full in the turbulent flow regime. The Hazen-Williams formula can be written as:

$$V = 0.85C_kR^{0.63}S^{0.54}$$

Also, $$Q = aV \text{ and } a = 4\pi R^2$$

Where:

V = Average Velocity of Fluid (m/sec.)

C_k= Hazen-Williams Friction Factor (no units)

R = Hydraulic Radius (m) ie cross sectional area of conduit divided by the wetted perimeter

S = Slope of hydraulic grade line (no unit).

Q = Quantity of flow or flow rate (m^3/s)

A = Cross sectional area of the pipe (m^2)

This empirical formula developed in the early 1900s by Allan Hazen and Gardner Williams is commonly used to compute the flow of water through pipes in water distribution systems, irrigation systems, fire sprinkler systems and sewer force mains.

NOTES:

• Hazen-Williams equation is valid for turbulent flow only.

• Equation and factors have been derived and are valid for water only. Properties for other fluids cannot be taken into account.

The importance of Hazen-Williams pipe design formula

In general, pipe design formulae play a crucial role in the field of civil engineering and hydraulics. They are used to determine the appropriate size, material, and layout of pipes to ensure the safe and efficient transport of fluids such as water, oil, or gas. These formulae take into account factors such as flow rate, pressure, pipe diameter, and friction loss to calculate the optimal design parameters for **a given system.**

N/B: This equation allows you to determine the required diameter for a given flow rate and other parameters, taking into account the frictional losses in the pipe. It's important to note that pipe diameter selection involves other considerations such as velocity limits, available pipe sizes, and system constraints. The Hazen-Williams formula provides a starting point for preliminary pipe sizing, but further analysis and design iterations may be necessary.

Application of Hazen-Williams formula to calculate diameter of pipes.

The Hazen-Williams formula can be rearranged to calculate the diameter of a pipe when the flow rate, roughness coefficient, pipe length, and slope are known. The equation can be written in terms of **d** (the diameter of the pipe). This equation allows you to determine the required diameter for a given flow rate and other parameters, taking into account the frictional losses in the pipe.

Standing pipe sizes.

The sizing of standing pipes can vary depending on the specific application, building codes, and engineering standards followed in different countries. However, find below, a general guideline for common standing pipe sizes used in plumbing systems. It's important to consult local codes, regulations, and professional engineering resources for accurate and up-to-date information specific to your location. Here are some commonly used standing pipe sizes:

1. **Residential Plumbing:**

- Supply Lines: 1/2 inch or 3/4 inch diameter (copper, PEX, or CPVC pipes).
- Branch Lines: 1/2 inch or 3/4 inch diameter (copper, PEX, or CPVC pipes).

2. **Commercial Plumbing:**

- Supply Lines: 1/2 inch to 2 inches diameter (depending on the water demand and building size).
- Branch Lines: 1/2 inch to 2 inches diameter (depending on the water demand and fixture requirements).

3. **Fire Protection Systems:**

- Sprinkler Systems: Typically range from 3/4 inch to 2 inches diameter (varies based on building occupancy, hazard classification, and fire codes).

- Standpipe Systems: Typically 2 inches or larger diameter (varies based on building height and fire codes).

These are general size ranges and may not cover all possible scenarios or specific requirements. It is crucial to consult local plumbing codes, building regulations, and professional engineers or plumbing experts for accurate sizing information based on the specific application and local guidelines.

Classification and rating of pipes.

The classification and rating of pipes can vary depending on the specific industry and application. However, below is a general overview of how pipes are classified and rated in various contexts.

1. **Material Classification:**

   Pipes can be classified based on the material they are made of. Common materials for pipes include:

- Metal Pipes: Such as steel, copper, cast iron, and stainless steel.
- Plastic Pipes: Including PVC (Polyvinyl Chloride), CPVC (Chlorinated Polyvinyl Chloride), PEX (Cross-linked Polyethylene), and HDPE (High-Density Polyethylene).
- Concrete Pipes: Used in sewer and drainage systems.
- Composite Pipes: Made from a combination of materials, such as fiber-reinforced plastic (FRP) pipes.

2. **Pressure Rating:**

   Pipes are also rated based on their pressure-carrying capacity. The pressure rating determines the maximum pressure the pipe can handle without failure. Pressure ratings are typically specified in pounds per square inch (psi) or bars. For example, in the case of plastic pipes, you might see pressure ratings like "PN10" or "150 psi" indicating the maximum working pressure.

3. **Size Classification:**

   Pipes are categorized based on their nominal diameter (often abbreviated as "NPS" or "DN") which represents the approximate inside diameter of the pipe. Common size classifications include:

- Standard Pipe Sizes: Follow industry standards such as the American National Standards Institute (ANSI) for steel pipes or the National Pipe Thread (NPT) standard for threaded pipes.
- Metric Pipe Sizes: Commonly used in countries that follow the metric system, such as Europe and some parts of Asia.

Water hammer,

Water hammer is a hydraulic phenomenon that occurs in piping systems when there is a sudden change in the flow rate or velocity of water. It leads to pressure surges and rapid pressure fluctuations that can cause damage to pipes, valves, and other system components. Understanding the causes and taking preventive measures can help mitigate water hammer.

Causes of Water Hammer:

1. Sudden Valve Closure: When a valve is closed quickly, it abruptly stops the flow of water, causing a pressure wave to travel back through the piping system.
2. Pump Start/Stop: The starting or stopping of pumps can result in sudden changes in flow rate, leading to water hammer.
3. Rapid Opening/Closing of Valves: Quick operation of valves, especially large ones, can cause water hammer due to sudden changes in flow velocity.
4. Air in the System: The presence of air pockets or trapped air in the piping system can compress and release, generating water hammer.
5. Water Column Separation: If the flow is not properly regulated, it can result in water column separation, leading to water hammer during rejoining.

Prevention of Water Hammer:

1. Surge Suppressors: Surge suppressors, also known as water hammer arrestors or expansion tanks, are devices that absorb and dissipate the pressure surge caused by water hammer, reducing its impact on the system. They provide a cushion to absorb the sudden changes in pressure.
2. Slow Valve Closure: Ensure that valves are closed gradually rather than abruptly to minimize the pressure surge.

3. Check Valves and Surge Relief Valves: Installing check valves helps prevent backflow and water column separation. Surge relief valves are designed to relieve excess pressure during water hammer events.
4. Air Release and Vacuum Breaker Valves: Properly designed air release valves and vacuum breaker valves help eliminate air pockets and prevent excessive air compression or vacuum formation.
5. Flow Control Devices: The use of flow control devices like throttling valves or flow restrictors can help regulate the flow velocity and reduce the likelihood of water hammer.
6. Pipe Support and Anchoring: Properly supporting and anchoring the pipes can minimize vibrations and movements that contribute to water hammer.
7. System Design Considerations: Proper hydraulic design, including pipe sizing, pressure relief measures, and gradual changes in pipe diameter or direction, can help reduce the occurrence of water hammer.

The use of air vessels, surge tanks and relief valves on water supply systems

Air vessels, surge tanks, and relief valves are important components used in water supply systems to manage pressure surges, control water hammer, and maintain system stability. Here's an explanation of their use:

1. **Air Vessels** (Air Chambers): Air vessels, also known as air chambers, are installed in water supply systems to provide cushioning and absorb pressure surges caused by water hammer. They are typically a vertical pipe section with a trapped air pocket located near the water source or at strategic points in the piping system. The air in the vessel compresses when there is a surge in pressure, and then gradually releases the stored energy to prevent sudden pressure fluctuations and pipe damage. Air vessels help reduce the effects of water hammer and provide surge protection.
2. **Surge Tanks:** Surge tanks are large-capacity reservoirs or tanks installed in water supply systems to absorb pressure fluctuations caused by rapid changes in flow rates. They are usually located at high points in the system or downstream of pumps or control valves. Surge tanks allow excess water to be temporarily stored, reducing pressure surges and controlling system stability.

They act as a buffer, accommodating sudden changes in flow and minimizing the impact on pipes, pumps, and other system components.

3. **Relief Valves**: Relief valves are safety devices installed in water supply systems to protect against excessive pressure build-up. They are designed to open and release water when the pressure exceeds a predetermined threshold, preventing damage to the system. Relief valves help maintain safe operating conditions by allowing the controlled release of excess pressure during abnormal conditions, such as water hammer events, pump failure, or pressure surges. They ensure system protection and prevent catastrophic failures.

The combined use of air vessels, surge tanks, and relief valves helps to control pressure fluctuations and water hammer, ensuring the safety and longevity of water supply systems. Proper sizing, installation, and maintenance of these components are essential for their effective operation.

Valves used in water works;

(a) Pressure Relief Valves:

- Pressure relief valves are essential safety devices used to protect water supply systems from overpressure situations. They automatically release excess pressure from the system when it exceeds a pre-set limit. This prevents potential damage to the pipeline, fittings, and other components. Pressure relief valves typically consist of a spring-loaded mechanism that opens when the internal pressure reaches a critical point, allowing water to escape and return the pressure to a safe level.

(b) Stop Valve:

- Stop valves, also known as isolation valves or shut-off valves, are used to completely stop or start the flow of water in a pipeline. They have a simple on/off mechanism and are typically used to isolate a specific section of the water supply system for maintenance, repairs, or emergencies. Stop valves are commonly found in homes to shut off the water supply to individual fixtures like sinks, toilets, and showers.

(c) Flow Regulation Valves:

- Flow regulation valves, as the name suggests, are used to control the rate of water flow through the pipeline. These valves allow for precise adjustments of flow rates, making them suitable for applications where flow control is essential. One common type of flow regulation valve is the globe valve, which uses a movable disk or plug to control the flow by varying the opening size.

(d) Pressure Regulation Valves:

- Pressure regulation valves, also known as pressure-reducing valves (PRVs), are designed to maintain a consistent and safe water pressure in a section of the water supply system. They are installed to reduce high incoming water pressure from the main supply to a lower, more manageable pressure for domestic or industrial use. Pressure regulation valves help prevent damage to appliances, fixtures, and pipes that could occur due to excessive pressure.

These various types of valves play crucial roles in maintaining the stability, safety, and efficiency of water supply systems. They are selected and installed based on the specific requirements and characteristics of the system they serve.

Operational details of the various types of valves explained above

(a) Pressure Relief Valve:

- Example: Watts 530C Pressure Relief Valve
- Operational Details: The Watts 530C Pressure Relief Valve is a commonly used valve in water supply systems. It consists of a spring-loaded mechanism and a disc or piston. The valve is installed in a pipeline or container where it continuously monitors the pressure. If the pressure exceeds a pre-set limit (e.g., due to thermal expansion or a system malfunction), the spring compresses, causing the valve to open. As the valve opens, excess water is released, lowering the pressure back to a safe level. Once the pressure is reduced, the spring reseats the disc or piston, closing the valve again.

(b) Stop Valve:

- Example: Gate Stop Valve
- Operational Details: The Gate Stop Valve is a type of stop valve commonly used in water supply systems. It features a gate-like barrier (made of metal or plastic) that slides up and down inside the valve body. When the valve handle is turned clockwise, the gate is lowered into the water flow path, effectively stopping the flow. Conversely, turning the handle counterclockwise lifts the gate, allowing water to pass through. The gate provides a full open or full closed position, making it ideal for isolating specific sections of the pipeline during maintenance or emergencies.

(c) Flow Regulation Valve:

- Example: Globe Valve
- Operational Details: The Globe Valve is a widely used flow regulation valve in water supply systems. It consists of a movable disk or plug and a stationary ring seat. When the valve handle is turned, the disk or plug moves toward or away from the seat, controlling the flow of water. In the fully open position, the disk or plug is lifted, allowing maximum flow. As the handle is turned to the closed position, the disk or plug presses against the seat, gradually reducing the flow until it is fully shut off. Globe valves provide precise flow control and are commonly used in applications where throttling is required.

(d) Pressure Regulation Valve:

- Example: Zurn Wilkins 70XLDU Pressure Reducing Valve
- Operational Details: The Zurn Wilkins 70XLDU Pressure Reducing Valve is a popular pressure regulation valve used in water supply systems. It is typically installed downstream from the main water supply to reduce high incoming pressure. The valve operates by automatically adjusting the size of an orifice based on the downstream pressure. When the downstream pressure rises above the setpoint, the valve's diaphragm and spring mechanism respond by constricting the orifice, reducing the water flow and, consequently, the pressure. Conversely, if the downstream pressure falls below the setpoint, the valve opens the orifice to allow more water flow and raise the pressure to the desired level.

These examples showcase the practical applications and operational principles of the different types of valves used in water supply systems, ensuring efficient and safe water distribution.

Constructional details of the valves described above

(a) Pressure Relief Valve - Watts 530C Pressure Relief Valve:

Constructional Details:

- Body: The valve body is typically made of brass or stainless steel to provide strength and corrosion resistance.
- Spring: Inside the valve, there is a spring that exerts force on the disc or piston to keep it closed when the pressure is within the desired range.
- Disc or Piston: The disc or piston is the movable component that seals the valve. It is designed to lift off its seat when the pressure exceeds the setpoint, allowing excess water to escape.

- Seat: The seat is a sealing surface against which the disc or piston closes to stop the flow when the valve is in its normal closed position.
- Outlet/Discharge: This is the port through which the excess water is released when the valve opens due to high pressure.
- Adjustment Mechanism (optional): Some pressure relief valves may have an adjustment mechanism to set the desired pressure limit for valve activation.

(b) Stop Valve - Gate Stop Valve:

Constructional Details:

- Body: The body of a gate stop valve is typically made of brass, bronze, or ductile iron to ensure durability and resistance to corrosion.
- Gate: The gate is a flat or wedge-shaped component that moves up and down within the valve body to control the water flow.
- Stem: The stem is connected to the gate and extends outside the valve body. It is manipulated by the valve handle to raise or lower the gate.
- Packing: To prevent water leakage around the stem, a packing material, such as Teflon or graphite, is used to create a watertight seal.
- Bonnet: The bonnet is the top cover of the valve body, securing the stem and providing access to the internal components during maintenance.
- Handle: The handle is attached to the stem and is used to manually operate the valve by turning it clockwise or counterclockwise.

(c) Flow Regulation Valve - Globe Valve:

Constructional Details:

- Body: Globe valve bodies are typically made of cast iron, bronze, or stainless steel to provide strength and resistance to corrosion.
- Disk or Plug: The disk or plug is a movable component that controls the flow of water by moving closer or away from the seat.
- Seat: The seat is a fixed, smooth, and flat sealing surface inside the valve body against which the disk or plug makes contact when closing.
- Stem: The stem connects the disk or plug to the valve handle and is responsible for transmitting the movement to control the flow.

- Bonnet: Similar to the gate stop valve, the bonnet covers the stem and provides access for maintenance and adjustments.
- Handwheel: The handwheel is a large wheel-like handle connected to the stem. It allows for easy manual operation of the valve.

(d) Pressure Regulation Valve - Zurn Wilkins 70XLDU Pressure Reducing Valve:

Constructional Details:

- Body: The body of a pressure reducing valve is commonly made of brass or bronze to ensure sturdiness and resistance to corrosion.
- Diaphragm: The diaphragm is a flexible, rubber-like component that responds to changes in downstream pressure to adjust the valve's orifice size.
- Spring: The spring inside the valve exerts force on the diaphragm, balancing the downstream pressure against the adjustable setpoint.
- Adjustment Screw: The pressure regulation setpoint can be adjusted using an adjustment screw, which compresses or releases the spring tension.
- Orifice: The orifice is an opening controlled by the movement of the diaphragm. When the downstream pressure exceeds the setpoint, the orifice size is reduced, and vice versa when the pressure drops below the setpoint.

These constructional details provide an overview of the key components and materials used in each type of valve, helping to understand their functions and how they operate in water supply systems.

The uses including practical areas of applications of the valves discussed above

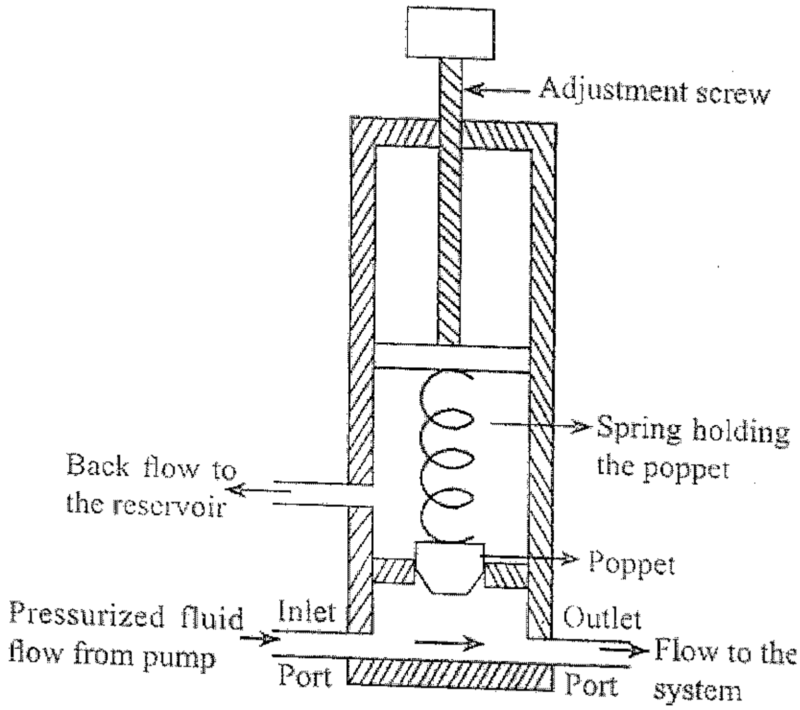
(a) Pressure Relief Valve - Watts 530C Pressure Relief Valve:

Use and Applications:

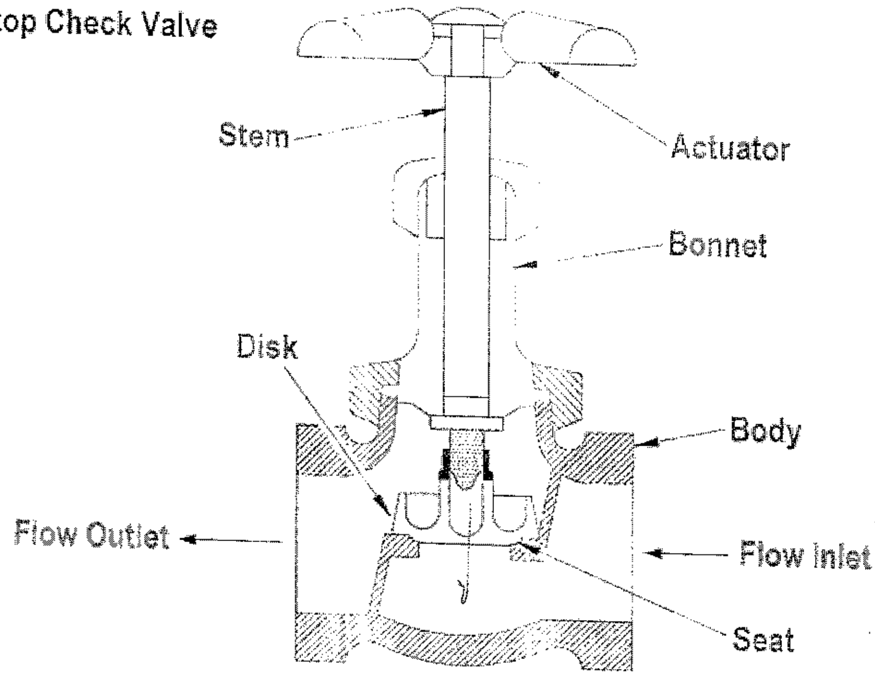
- Pressure relief valves are used in water supply systems to protect against overpressure situations. They are essential safety devices commonly installed in various locations, such as water heaters, boilers, and large storage tanks.
- Practical Applications: In a water heater, the pressure relief valve ensures that the pressure inside the tank does not exceed a safe limit. If the thermostat malfunctions or the water overheats, the valve will open and

VALVES FOR WATER WORKS

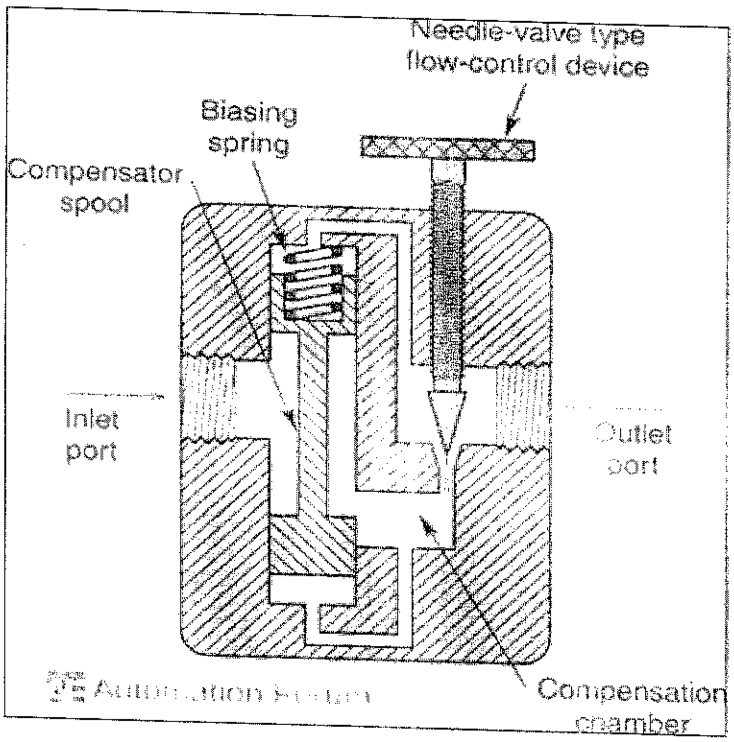
1 . PRESSURE RELIEF VALVE DIAGRAM



2 Stop Check Valve

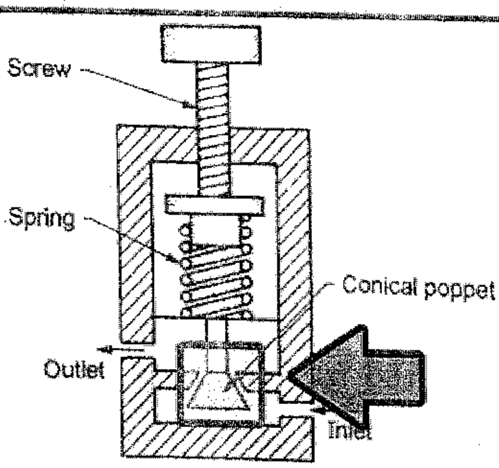


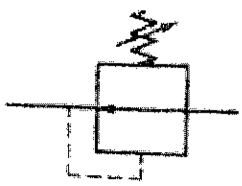
3 FLOW CONTROL VALVE DIAGRAM



4.

Pressure Reducing Valve





Symbol

discharge excess water to prevent a potential explosion. Similarly, in a large water storage tank, the valve prevents damage caused by excessive pressure during filling or thermal expansion.

(b) Stop Valve - Gate Stop Valve:

Use and Applications:

- Stop valves, particularly gate stop valves, are used to control the flow of water and to isolate specific sections of a water supply system. They provide a full open or full closed position, making them suitable for applications where on/off control is needed.
- Practical Applications: Gate stop valves are commonly found in residential plumbing systems. They are used to shut off the water supply to individual fixtures, such as sinks, toilets, and showers. In commercial or industrial settings, they are used to isolate sections of the water supply for maintenance or emergency repairs.

(c) Flow Regulation Valve - Globe Valve:

Use and Applications:

- Flow regulation valves, like globe valves, are designed to provide precise control over the flow rate of water. They are used in applications where throttling and fine-tuned adjustments of flow are necessary.
- Practical Applications: Globe valves are commonly used in industrial settings, such as in water treatment plants, where accurate control of water flow is crucial. They are also used in residential plumbing systems for controlling the flow of water to showers, faucets, and appliances like washing machines.

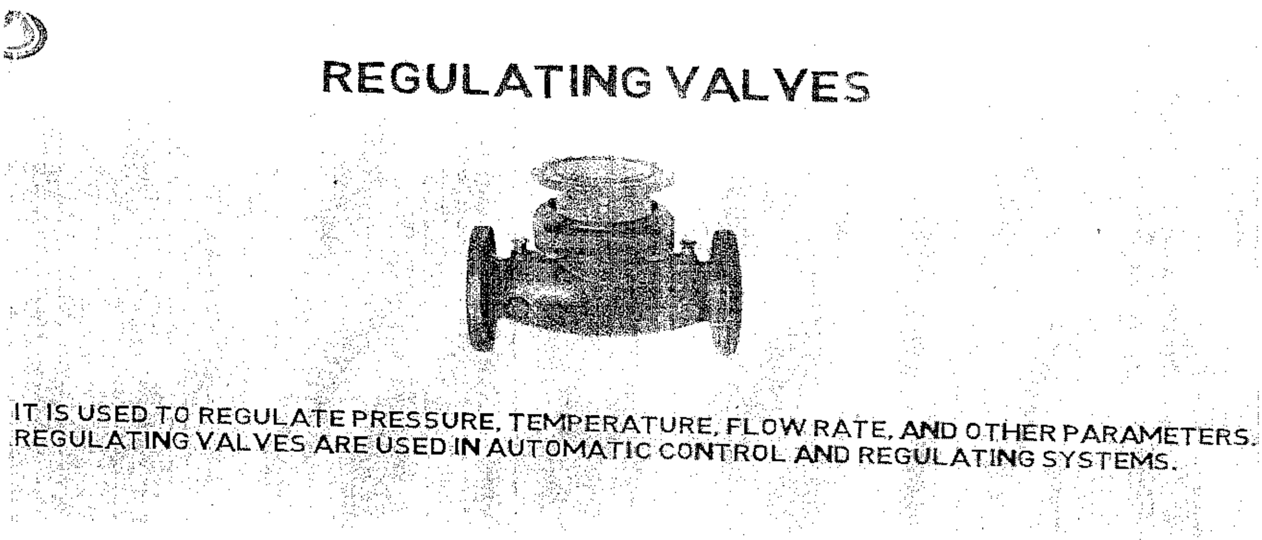
(d) Pressure Regulation Valve - Zurn Wilkins 70XLDU Pressure Reducing Valve:

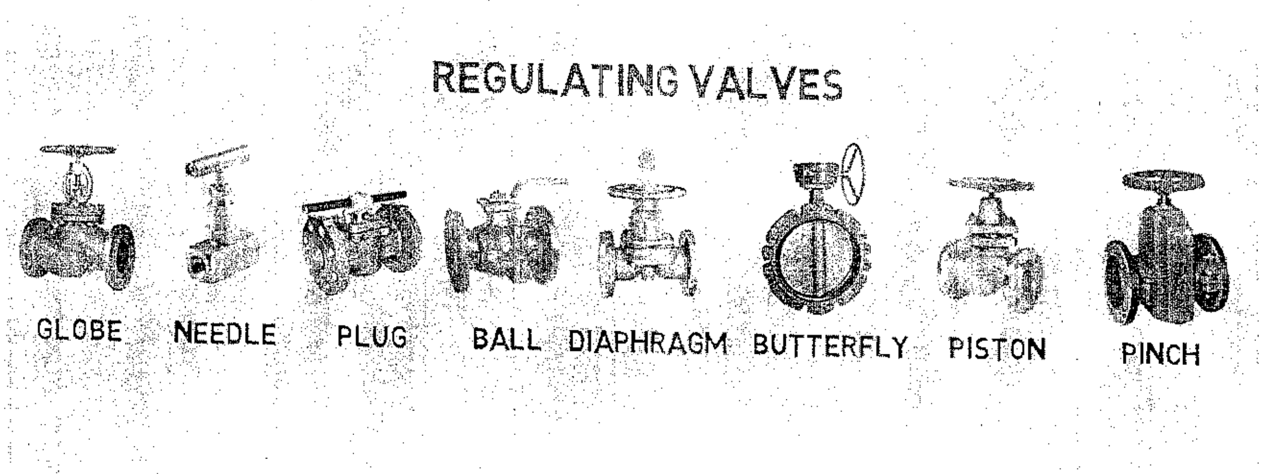
Use and Applications:

- Pressure regulation valves, specifically pressure reducing valves, are used to maintain a consistent and safe water pressure downstream from the main supply. They ensure that the pressure remains within an acceptable range for various applications.
- Practical Applications: Pressure reducing valves are widely used in residential, commercial, and industrial settings. In residential applications, they are installed to reduce the high pressure from the municipal supply to a safe level for domestic use. In commercial buildings and industries, pressure reducing valves are used to protect plumbing fixtures, appliances, and sensitive equipment that may be sensitive to high water pressures.

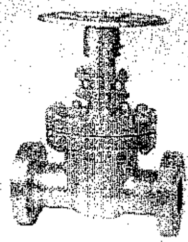
Overall, these valves play critical roles in maintaining the stability, safety, and efficiency of water supply systems in various practical applications. They are selected and installed based on the specific requirements of each system and the need for pressure regulation, flow control, or isolation.

PICTURES OFVARIOUS TYPES OF VALVES USED IN WATER WORKS



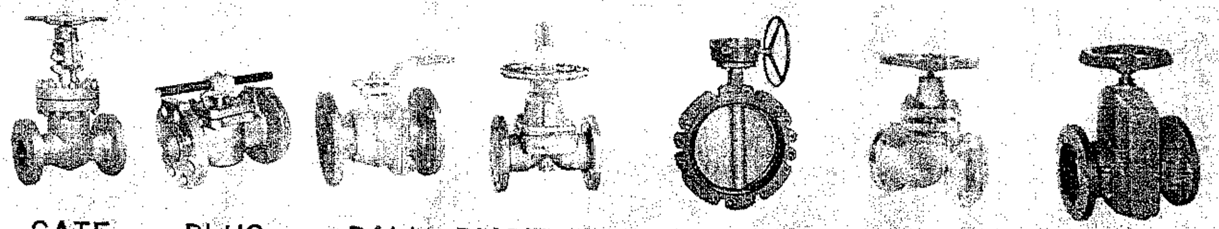


ISOLATING VALVES



THIS IS USED TO CONTROL THE FLOW. USUALLY FOR MAINTENANCE OR SAFETY PURPOSES.
IT IS INSTALLED WITHIN A PROPERTY IS REQUIRED TO BE IN AN ACCESSIBLE LOCATION.
THE LOCATION OF AN ISOLATING VALVE CAN VARY BASED ON THE TYPE AND LOCATION.

ISOLATING VALVES

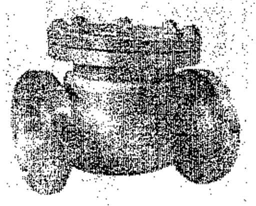


GATE PLUG BALL DIAPHRAGM BUTTERFLY PISTON PINCH

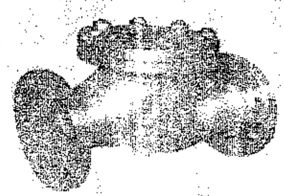
NON-RETURN VALVES

It prevents reversal of flow in the piping systems

NON-RETURN VALVES



SWING CHECK VALVE



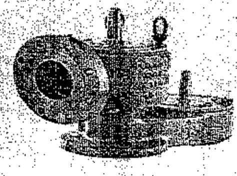
LIFT CHECK VALVE

Safety Relief Valve

A Relief valve used to control or limit the pressure in Piping s

Safety Relief Valve

Pressure Relief valve



Vaccume Relief valve

VALVES FOR SPECIAL PURPOSES

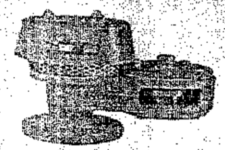
MULTI-PORT

FLUSH BOTTOM

FLOAT

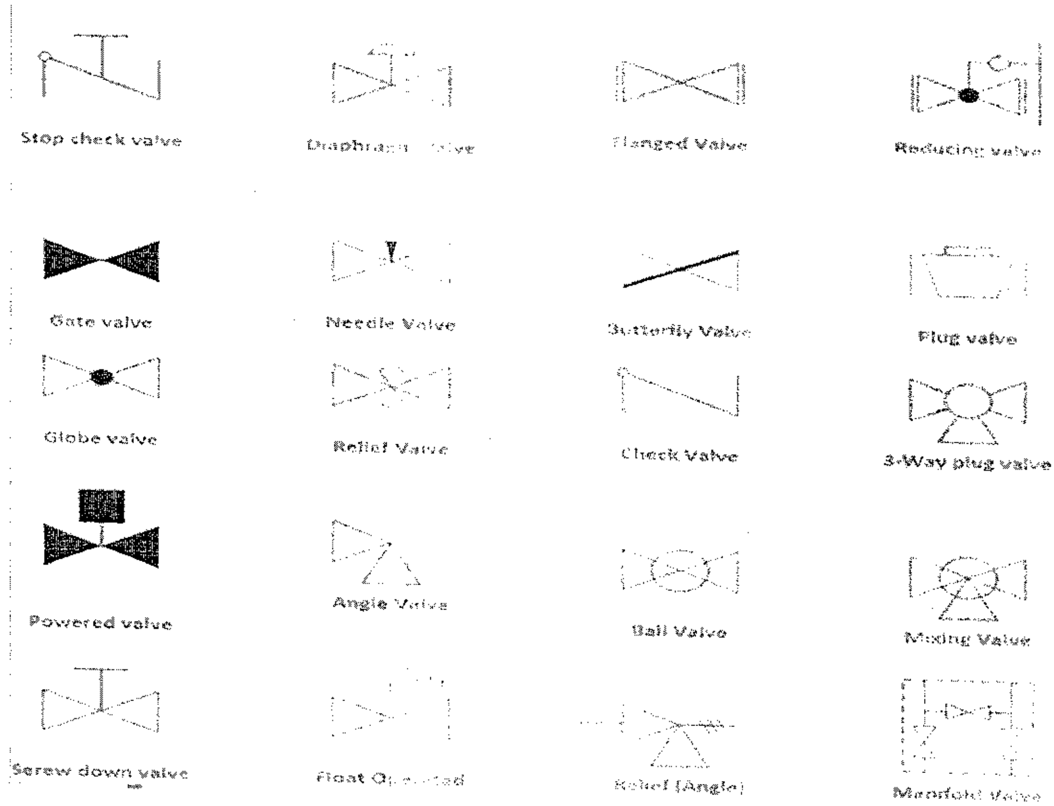
FOOT

PRESSURE RELIEF



BREATHER

REPRESENTATION OF VALVES ON DRAWINGS


Stop check valve
Diaphragm Valve
Flanged Valve
Reducing valve
Gate valve
Needle Valve
Butterfly Valve
Plug valve
Globe valve
Relief Valve
Check Valve
3-Way plug valve
Powered valve
Angle Valve
Ball Valve
Mixing Valve
Screw down valve
Float Operated
Relief (Angle)
Manifold Valve

3

MEC 424 - PIPING DESIGN LECTURE NOTES

Types of joints; solder or wiped, sweet, slipe, bell and spigot, threaded or screwed joints

Different types of joints provide various options for connecting pipes and fittings, offering flexibility, strength, and ease of installation in plumbing and piping systems. The specific joint used depends on factors such as the type of material, system requirements, intended application, and industry standards. Some of these joints include;

1. Solder or Wiped Joints:
    - Solder joints, also known as wiped joints, are commonly used to join copper pipes.
    - The joint is created by applying heat to the copper fittings and pipe, melting solder, and allowing it to flow into the gap between the fittings and pipe.
    - As the solder cools and solidifies, it forms a strong and watertight connection.
2. Sweat Joints:
    - Sweat joints are a type of solder joint used specifically for copper pipes.
    - The joint is created by heating the fittings and pipe using a torch, and then applying solder to the joint, which melts and flows into the gap between the fittings and pipe.
    - The name "sweat" comes from the process of heating the copper to the point where it appears to sweat before applying solder.
3. Slip Joints:
    - Slip joints, also known as compression joints, are commonly used in plumbing fixtures such as faucets and traps.
    - These joints involve a compression fitting that allows for easy assembly and disassembly without the need for soldering or welding.
    - The joint is formed by sliding the compression nut onto the pipe, followed by a compression ring or ferrule. The nut is then tightened, compressing the ring onto the pipe, creating a watertight seal.
4. Bell and Spigot Joints:
    - Bell and spigot joints are used in joining pipes that have a bell-shaped end and a corresponding spigot end.
    - The bell-shaped end of one pipe fits over the spigot end of another, creating a mechanical connection.

- These joints are commonly found in plastic pipes, such as PVC or ABS, and are often used in sewer, drainage, and underground applications.

5. Threaded or Screwed Joints:
    - Threaded joints involve pipes with threaded ends that can be screwed together to create a secure connection.
    - The joint is formed by threading the ends of the pipes and then using pipe thread sealant or tape to ensure a tight seal.
    - Threaded joints are commonly used in metal pipes and fittings, such as galvanized steel or stainless steel, and are often found in gas lines, water lines, and plumbing fixtures.

The use of each of the joints described above

Below is a breakdown of the use and application of each joint:

1. Solder or Wiped Joints:
    - Solder joints are commonly used in copper pipe systems for water supply, heating, and cooling applications.
    - They provide a secure and leak-free connection, particularly in residential and commercial plumbing systems.
    - Solder joints are widely used in potable water systems, including faucets, valves, and fittings.
2. Sweat Joints:
    - Sweat joints, also known as solder joints, are predominantly used in copper plumbing systems.
    - They are commonly found in residential and commercial water supply lines, both hot and cold.
    - Sweat joints are used in applications such as connecting copper pipes, fittings, and valves in potable water systems.
3. Slip Joints:
    - Slip joints are frequently used in plumbing fixtures that require easy assembly and disassembly, such as faucets, traps, and other drain connections.

- They provide a convenient way to connect and disconnect pipes without the need for soldering or specialized tools.
- Slip joints are commonly used in kitchen sinks, bathroom sinks, bathtubs, and other fixtures where periodic maintenance or replacement may be required.

4. Bell and Spigot Joints:
    - Bell and spigot joints are commonly used in plastic pipe systems, particularly for sewer, drainage, and underground applications.
    - They provide a simple and secure means of connecting pipes in these applications, ensuring a tight and watertight seal.
    - Bell and spigot joints are often used in municipal sewer lines, stormwater systems, and underground drainage pipes.
5. Threaded or Screwed Joints:
    - Threaded joints are widely used in various applications, including plumbing, gas lines, and industrial piping systems.
    - They provide a reliable connection that can withstand high pressures and temperatures.
    - Threaded joints are commonly found in water supply lines, natural gas lines, fire sprinkler systems, and hydraulic systems.

N/B. Each type of joint offers specific advantages based on the material being used, the application requirements, and the desired ease of assembly or disassembly. It's important to consider factors such as the type of piping material, system compatibility, pressure and temperature requirements, and local plumbing codes and standards when selecting the appropriate joint for a given plumbing or piping project.

Constructional details of each of the joints explained above

1. Solder or Wiped Joints:
    - Solder joints are created by applying heat to the copper fittings and pipe to melt the solder.
    - The joint is assembled by cleaning the surfaces of the fittings and pipe, applying flux to prevent oxidation, and then heating the joint with a torch.

- Once the joint is heated, solder wire or paste is applied to the joint, and it flows into the gap between the fittings and pipe through capillary action.
- As the joint cools, the solder solidifies and forms a strong and watertight connection.

2. Sweat Joints:
   - Sweat joints are a type of solder joint used for copper pipes.
   - The process begins by cleaning the fittings and pipe surfaces, applying flux to prevent oxidation, and heating the joint using a torch.
   - The heat causes the solder to melt and flow into the gap between the fittings and pipe.
   - The joint is typically heated until the solder evenly covers the gap and forms a smooth connection.
   - As the joint cools, the solder solidifies and creates a reliable and leak-free joint.
3. Slip Joints:
   - Slip joints consist of two main components: a compression nut and a compression ring or ferrule.
   - The joint is assembled by sliding the compression nut onto the pipe, followed by the compression ring or ferrule.
   - The pipe end is then inserted into the receiving fitting or connector.
   - The compression nut is tightened onto the fitting, compressing the ring or ferrule against the pipe.
   - The compression creates a seal, preventing leaks and ensuring a secure connection.
4. Bell and Spigot Joints:
   - Bell and spigot joints are commonly used in plastic pipes, such as PVC or ABS.
   - The bell-shaped end of one pipe fits over the spigot (plain, tapered) end of another pipe.
   - The joint relies on the interference fit between the bell and spigot to create a mechanical connection.
   - The joint may be further secured with adhesive or sealant specifically designed for plastic pipes.
   - The resulting connection provides a tight seal and structural integrity for sewer, drainage, and underground applications.
5. Threaded or Screwed Joints:
   - Threaded joints involve pipes with threaded ends that can be screwed together.

PIPING JOINTS

People also ask

1 What is a wiped joint?

The defining characteristic of a wiped joint is that the soldering process involves mechanically working or 'wiping' the joint. As well as heating solder and applying it to the joint, the solder is shaped into place manually, wiping it with a non-metallic tool to form a smooth-surfaced outer shape.

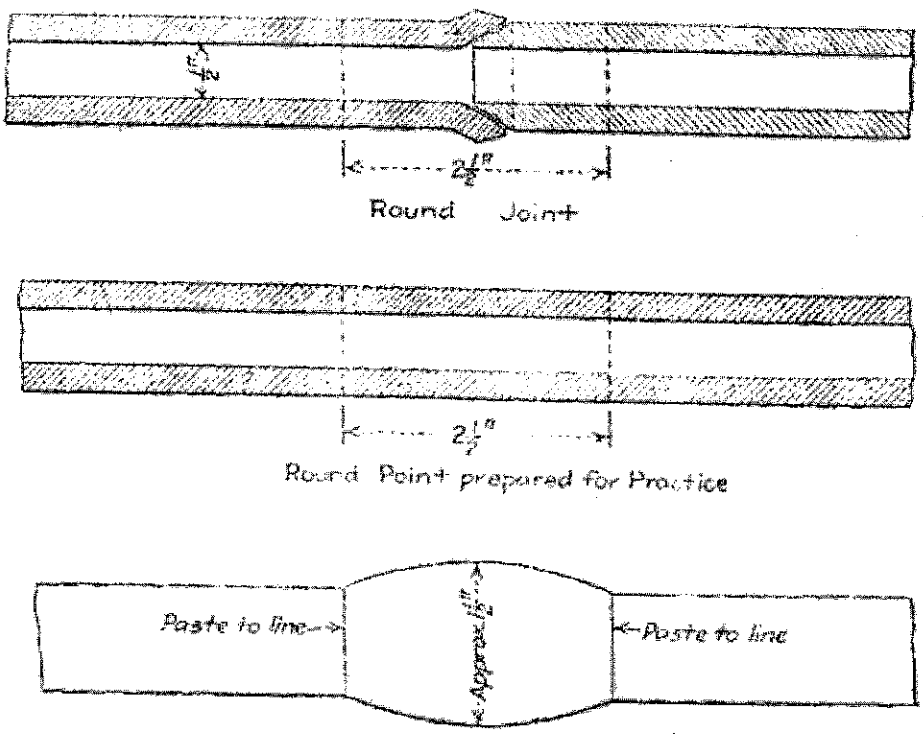


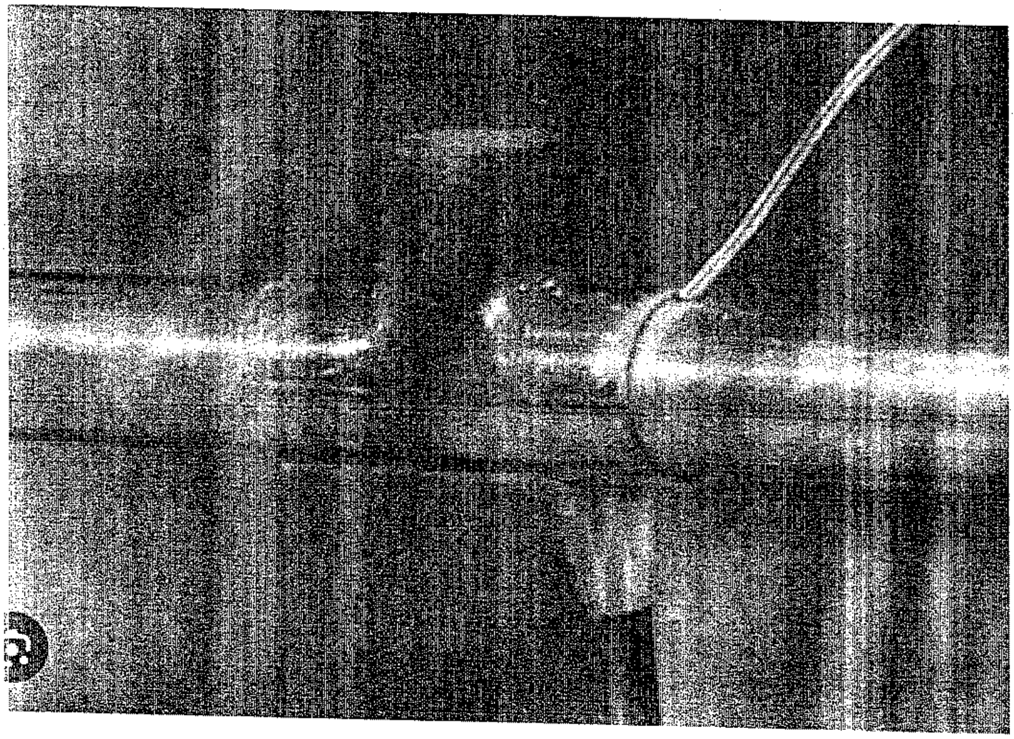
Fig. 23.

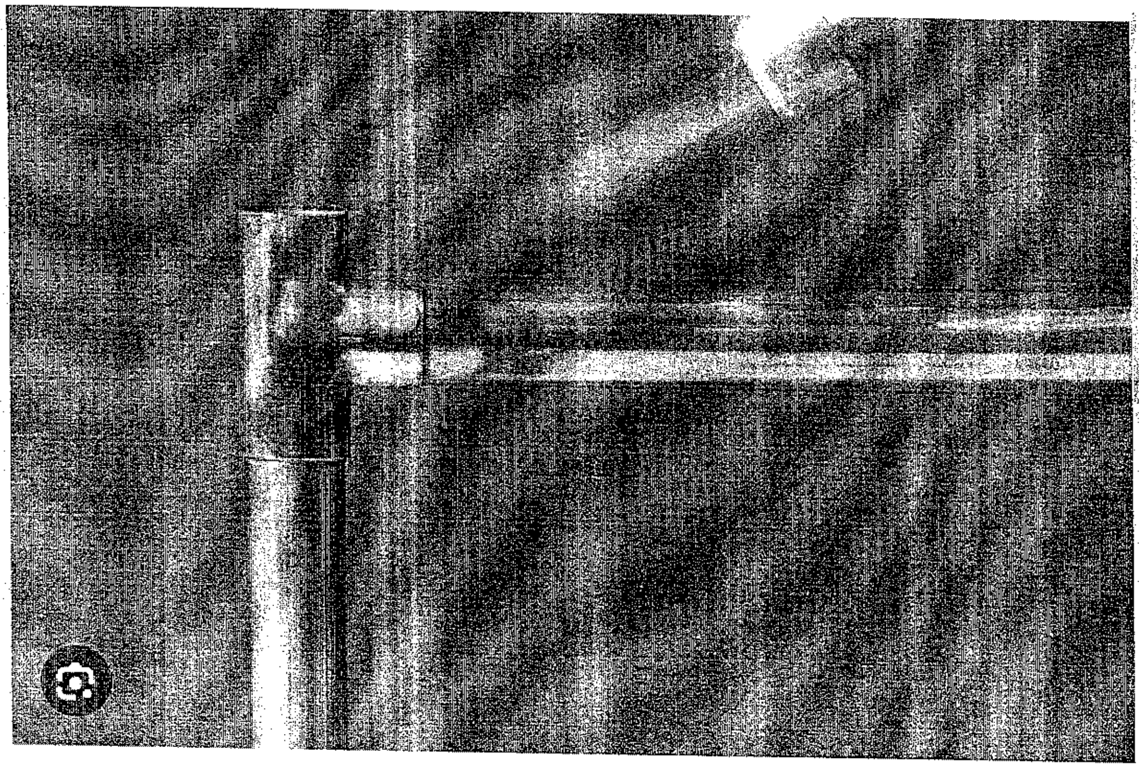
Half-Inch Round Wiped Joint

Visit

2. What is a sweat joint?

Soldering a pipe joint or "sweating a pipe" is accomplished by heating a copper fitting with a propane torch until the fitting is just hot enough to melt metal solder. The heat draws the solder into the gap between the fitting and pipe to form a water tight seal.

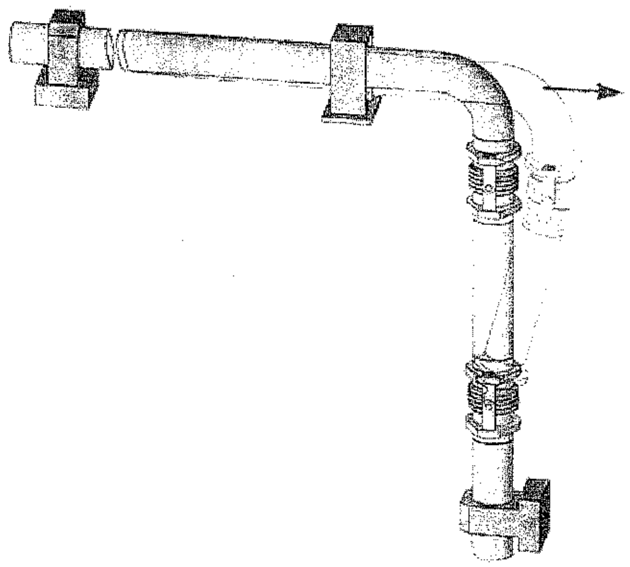




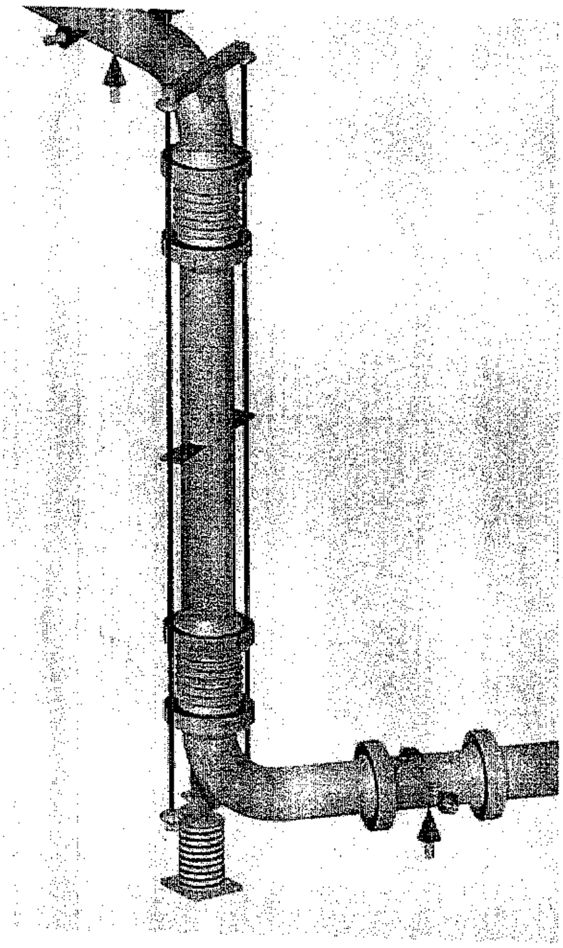
How to Sweat Copper Pipe (DIYer's Guide) - Bob Vila   Visit

3 What is a slip joint in piping?

A slip joint is a type of joint used in plumbing to connect pipes and fittings. It consists of a connecting nut and a gasket (often made of rubber or neoprene) that are placed between the end of one pipe and the fitting. The joint is then tightened with a wrench, compressing the gasket and forming a seal.



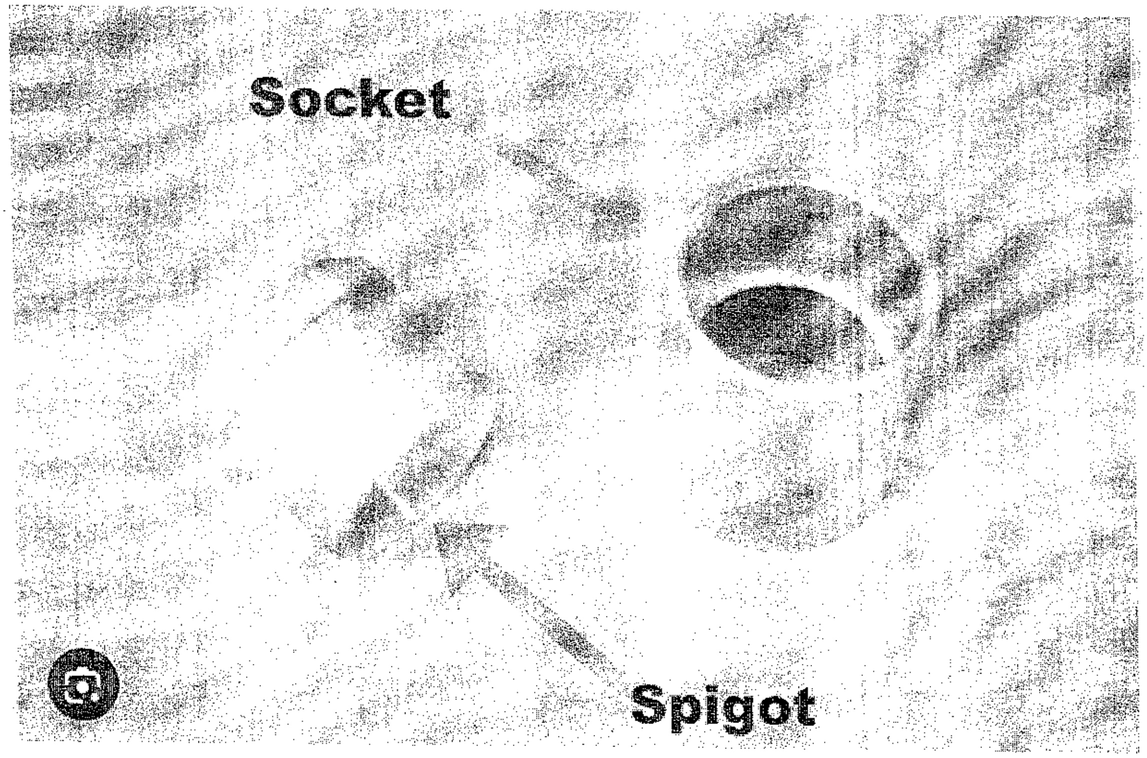
Pipe expansion bellows - Pipe expansion joints - Flexible joints for pipes - Spiroflex

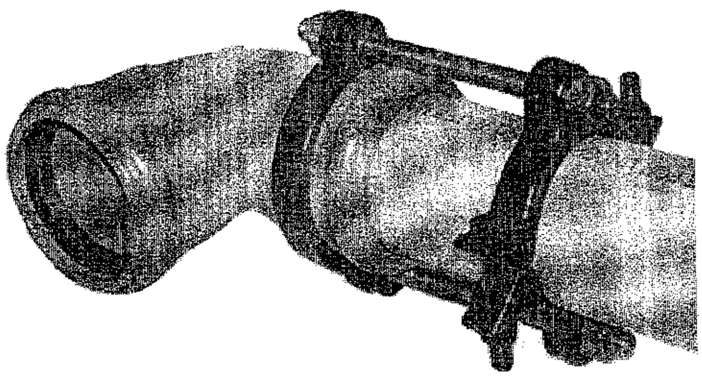


10 Expansion Joint Basics to Remember ... for Pipe Stress Analysts

4 What is bell and spigot joint?

A form of joint used on pipes that have an enlarged diameter or bell at 1 end and a spigot at the other that fits into and is laid in the bell.

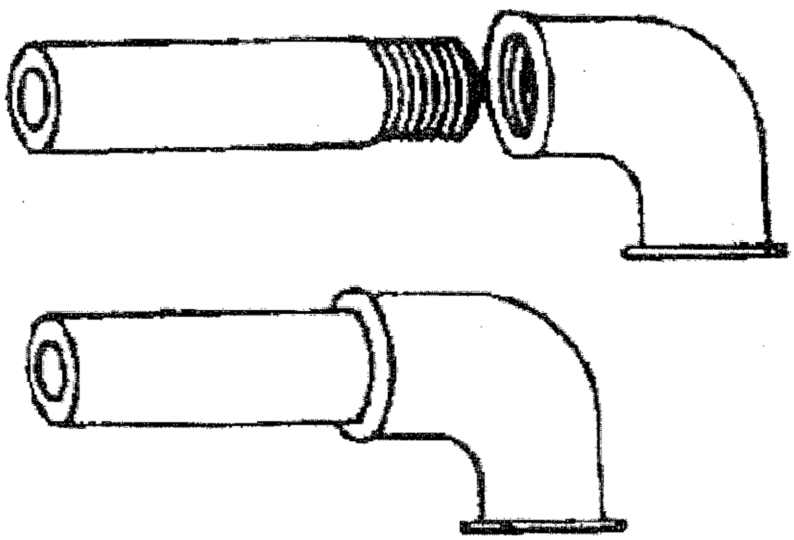




| Serrated Joint Restraint | Bell & Spigot PVC

5. SCREWED OR THREADED JOINTS

www.nzdl.org



Threaded Joints - Course: Techniques of Fitting and Assembling Component Parts to Produce Simple Units....

- The joint is created by threading the ends of the pipes using a die or threading machine.
- Pipe thread sealant or tape is applied to the threads to ensure a tight seal and prevent leaks.
- The threaded ends of the pipes are then screwed together, creating a secure and durable connection.
- Threaded joints are commonly used in metal pipes, such as galvanized steel or stainless steel, and can withstand high pressures and temperatures.

Typical joining methods for given installations as well as suitable supports for the pipes

1. Water Supply Systems (Non-Pressure):
    - Joining Methods: Soldering or brazing for copper pipes, solvent welding for PVC or CPVC pipes, and compression fittings for plastic or metal pipes.
    - Suitable Supports: Plastic or metal pipe hangers, adjustable pipe clamps, and pipe straps.
2. Water Supply Systems (Pressure):
    - Joining Methods: Threaded connections with pipe thread sealant or tape for metal pipes, press fittings for copper or PEX pipes, and fusion or mechanical joints for HDPE pipes.
    - Suitable Supports: Metal pipe hangers, clevis hangers, and strut channels with pipe clamps.
3. Gas Supply Systems:
    - Joining Methods: Threaded connections with pipe thread sealant or tape for metal pipes, compression fittings for copper or stainless steel pipes, and mechanical joints for PE or PVC pipes.
    - Suitable Supports: Metal pipe hangers specifically designed for gas piping, adjustable pipe clamps, and strut channels with pipe clamps.
4. Heating and Cooling Systems:
    - Joining Methods: Soldering or brazing for copper pipes, press fittings for copper or PEX pipes, and threaded connections for steel pipes.
    - Suitable Supports: Metal pipe hangers, adjustable pipe clamps, and pipe straps.
5. Drainage and Sewer Systems:

- Joining Methods: Solvent welding or rubber gasket joints for PVC or ABS pipes, bell and spigot joints for cast iron or clay pipes, and mechanical joints for ductile iron pipes.
- Suitable Supports: Pipe supports designed for drainage systems, such as pipe clamps, pipe rollers, and hangers with resilient inserts.

6. Fire Sprinkler Systems:
   - Joining Methods: Threaded connections with pipe thread sealant for steel pipes, groove connections for ductile iron pipes, and press fittings for copper or PEX pipes.
   - Suitable Supports: Metal pipe hangers with resilient inserts, strut channels with pipe clamps, and fire-rated hangers as required by local codes.

When selecting joining methods and supports, it is crucial to refer to the specific requirements of the installation, local codes and regulations, and manufacturer guidelines. These considerations will help ensure the integrity, safety, and compliance of the pipework system. Consulting with qualified professionals, such as plumbers or mechanical engineers, is highly recommended to determine the most suitable joining methods and supports for your specific installation.

Pipe fittings such as elbows, reducers tees etc

Pipe fittings are components used to connect and redirect pipes in a plumbing or piping system. They come in various shapes, sizes, and types to accommodate different needs and configurations. Some typical examples of these fittings include;

1. Elbow: Elbows are fittings that change the direction of a pipe. They are typically available in 45-degree and 90-degree angles, allowing pipes to be bent at those angles. Elbows help navigate around obstacles or create turns in the pipe system.
2. Tee: Tees have a T-shaped design, with one inlet and two outlets (or vice versa). They allow for a branch line to be connected at a right angle to the main pipe, creating a branching connection.
3. Reducer: Reducers are fittings used to connect two pipes of different sizes. They come in two types: concentric reducers, where the ends are aligned in a straight line, and eccentric reducers, where the ends are offset.
4. Coupling: Couplings are fittings used to join two pipes of the same size. They are typically used when a permanent connection is desired, and they provide a leak-resistant joint.

5. Union: Unions are similar to couplings, but they allow for the disconnection and reconnection of pipes without requiring the fittings to be cut or disassembled. Unions are often used when periodic maintenance or repairs are anticipated.
6. Cross: Cross fittings have a plus-shaped design with four connection points. They allow for the branching of pipes in multiple directions, typically at right angles to each other.
7. Flange: Flanges are flat, circular fittings with holes around the perimeter. They are used to connect pipes by bolting them together, creating a secure and leak-resistant joint. Flanges are commonly used in high-pressure or large-diameter piping systems.
8. Cap: Caps are fittings that are used to seal the end of a pipe. They are typically used for terminating a line or temporarily closing off a pipe for maintenance or testing purposes.

These are just a few examples of pipe fittings commonly used in plumbing and piping systems. It's important to note that there are many more types of fittings available, and each serves a specific purpose in a piping system. The selection of the appropriate fittings depends on factors such as pipe size, material, fluid type, system requirements, and local codes and standards.

Uses of fittings such as elbows, reducers, tees etc

Common uses of some popular pipe fittings:

1. Elbow:
    - Elbows are used to change the direction of pipe flow, typically by 45 degrees or 90 degrees.
    - They are crucial for routing pipes around obstacles, corners, or tight spaces.
    - Elbows help in aligning pipes for easier installation and maintenance.
2. Reducer:
    - Reducers are used to connect pipes of different sizes, allowing for a smooth transition in diameter.
    - They ensure compatibility between pipes with different dimensions.
    - Reducers help in adapting the flow rate and pressure between two pipe sections.
3. Tee:
    - Tees are used for creating branching connections in a pipeline.

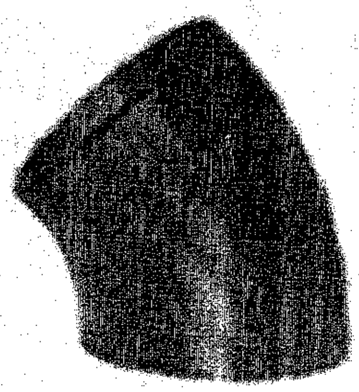
- They provide an outlet at a 90-degree angle to the main pipe. allowing for a separate line to be connected.
- Tees are commonly used in applications such as distributing water supply to multiple fixtures or creating parallel lines in a system.

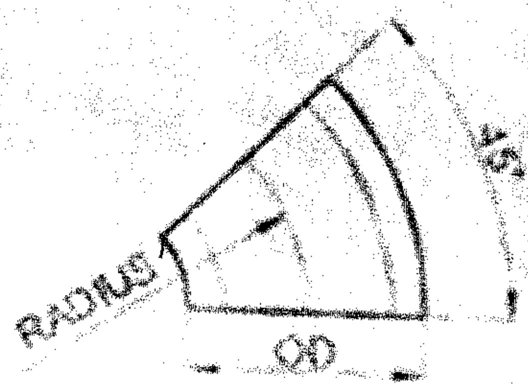
4. Coupling:
   - Couplings are used to join two pipes of the same size together.
   - They provide a secure and leak-resistant connection.
   - Couplings are commonly used for permanent installations where a strong and permanent joint is required.
5. Union:
   - Unions are used for easily disconnecting and reconnecting pipes without the need for cutting or disassembling the fittings.
   - They provide a convenient access point for maintenance or repairs.
   - Unions are commonly used in systems that may require periodic disconnection, such as pump connections or equipment servicing.
6. Cross:
   - Cross fittings allow branching in multiple directions, typically at right angles to each other.
   - They are used to create intersecting lines in complex pipe systems.
   - Cross fittings enable the distribution of fluid flow in various directions.
7. Flange:
   - Flanges are used to connect pipes by bolting them together.
   - They provide a strong and secure joint, particularly in high-pressure and large-diameter piping systems.
   - Flanges are commonly used in industries such as oil and gas, chemical processing, and power generation.
8. Cap:
   - Caps are used to seal the end of a pipe, blocking the flow.
   - They are typically used for capping off unused or terminated pipe ends.
   - Caps are useful for temporarily closing a pipe for maintenance or testing purposes.

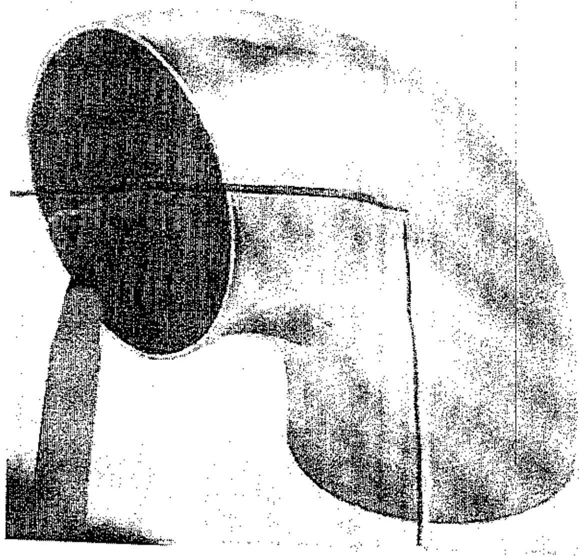
COUPLINGS

1 ELBOWS (90° and 45°)

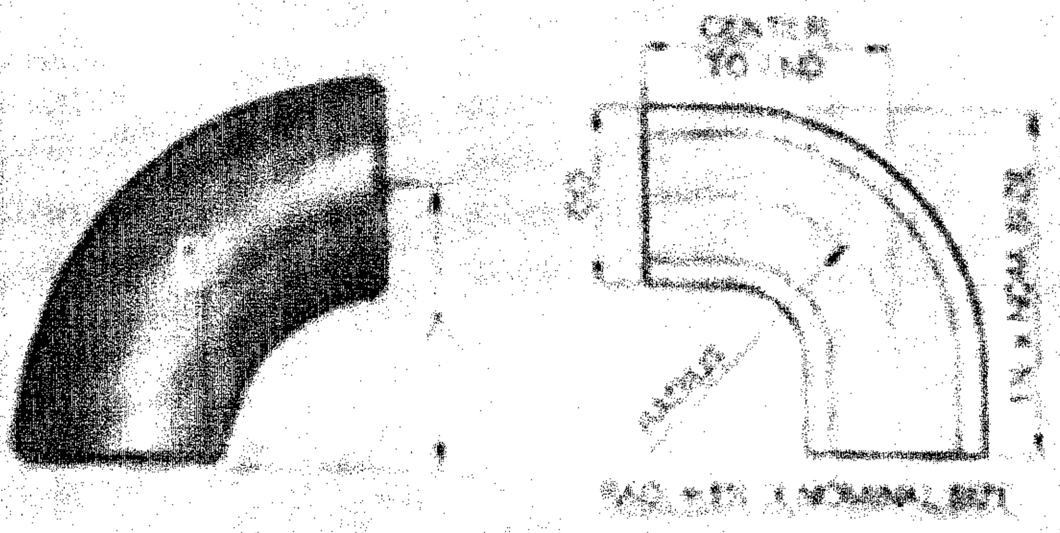
45° ELBOW



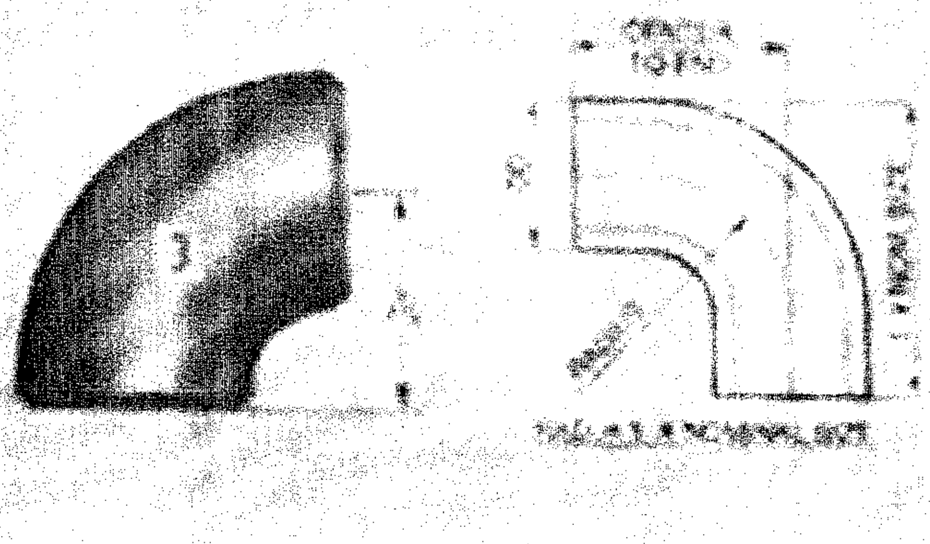




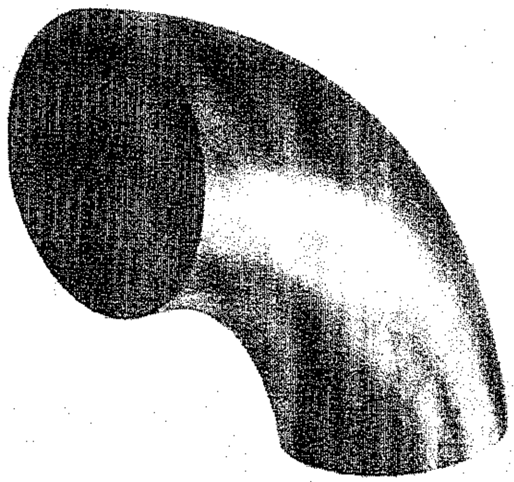
LONG-RADIUS ELBOW

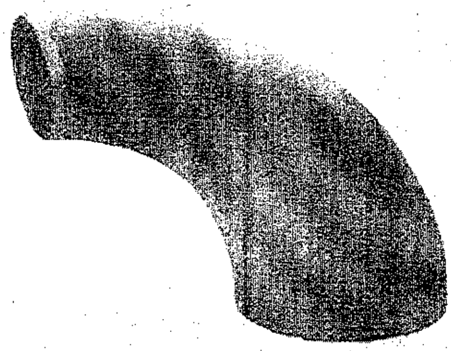


SHORT-RADIUS ELBOW

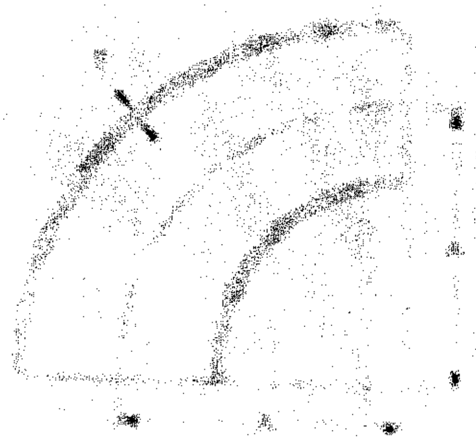


2 REDUCERS (90° and 45°)

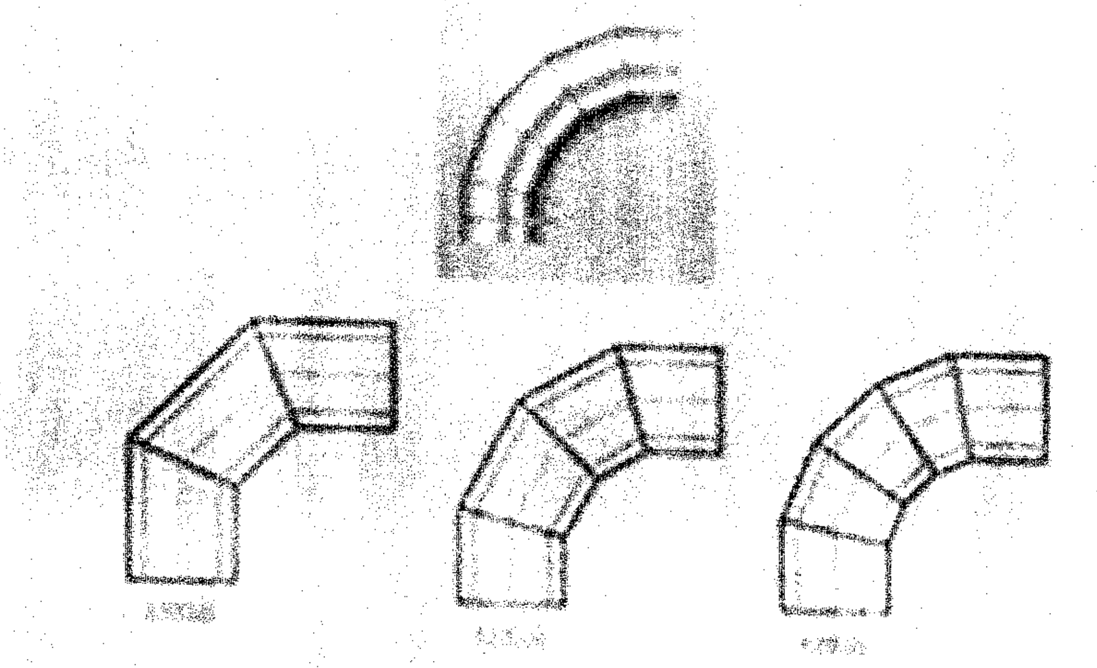


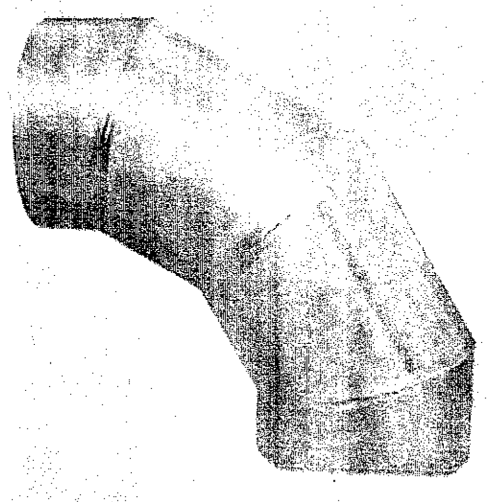


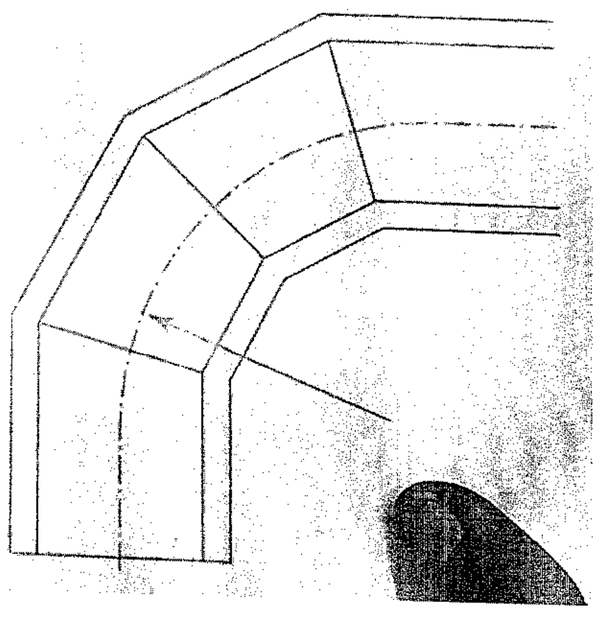
REDUCING ELBOWS

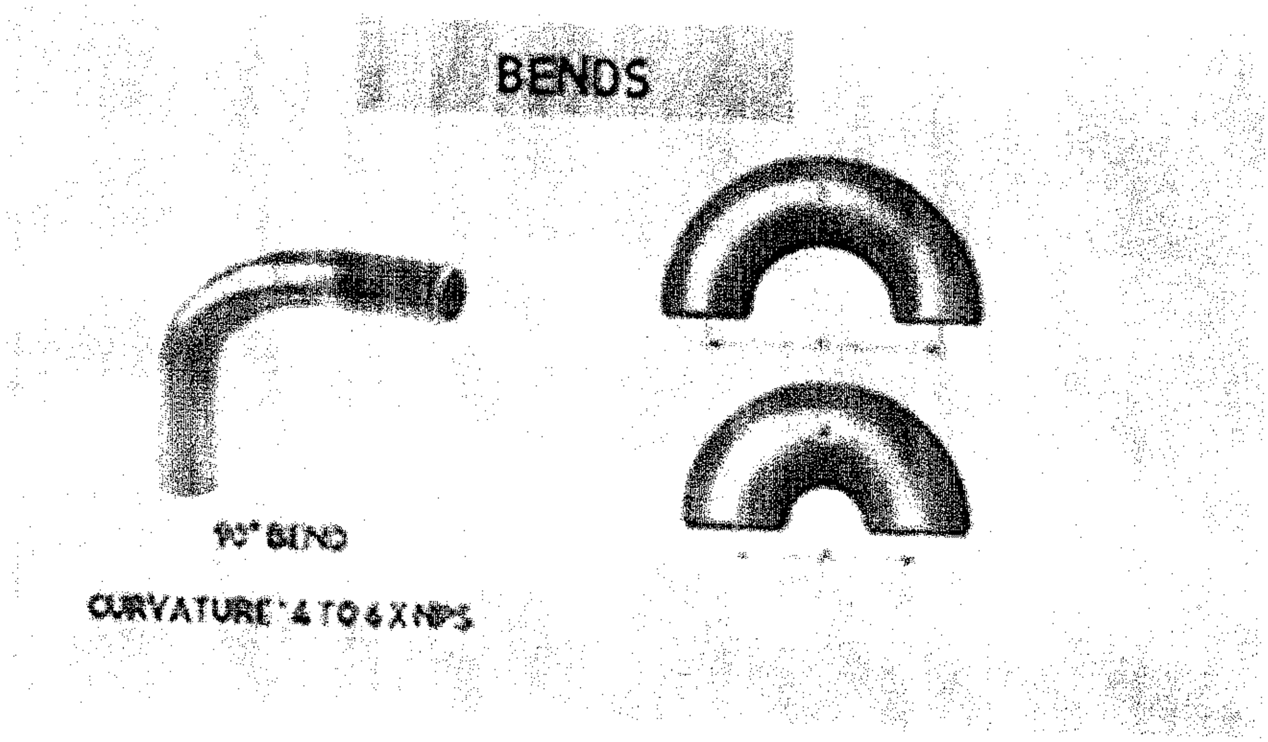


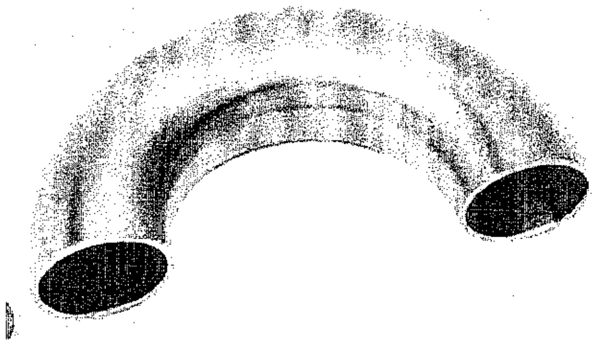
MITERED ELBOWS

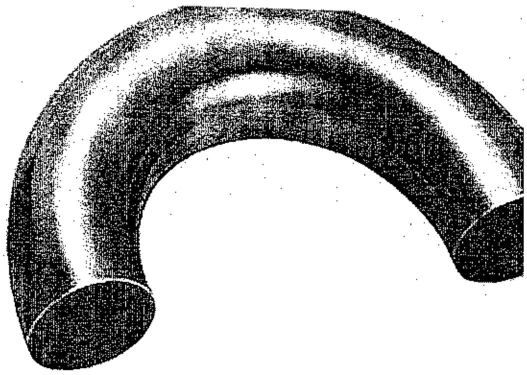




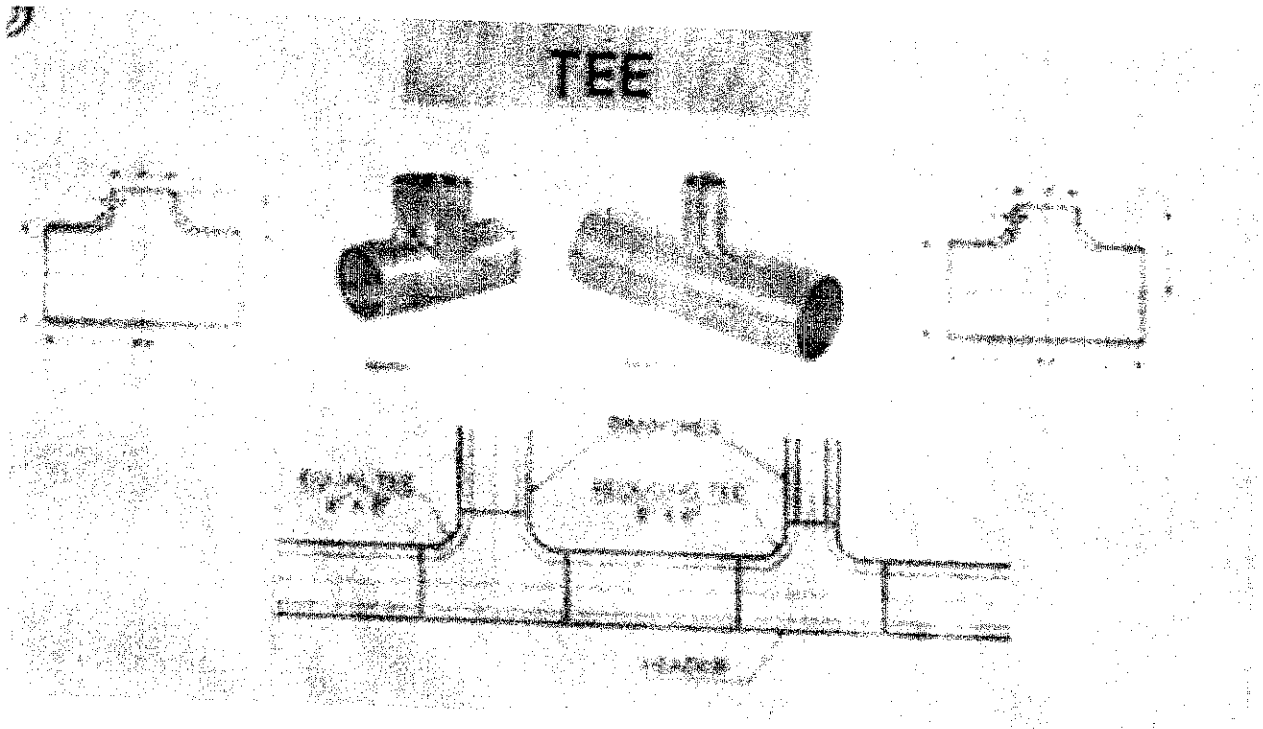


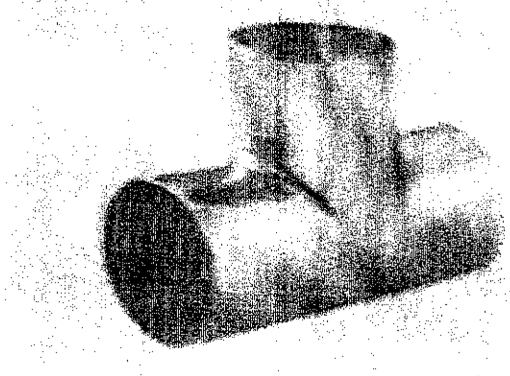


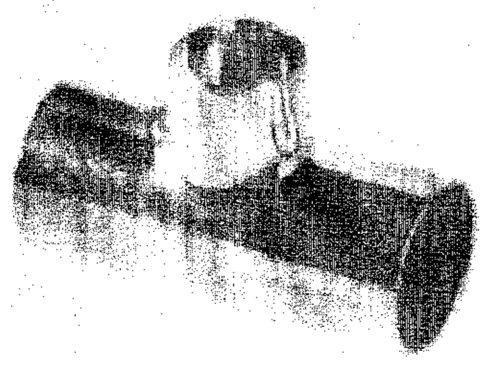




Sometimes they are called RETUNS (180°)



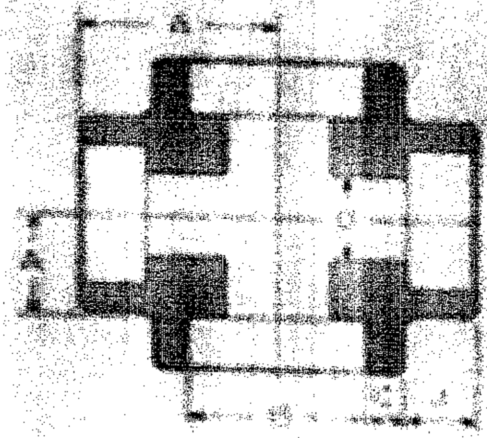


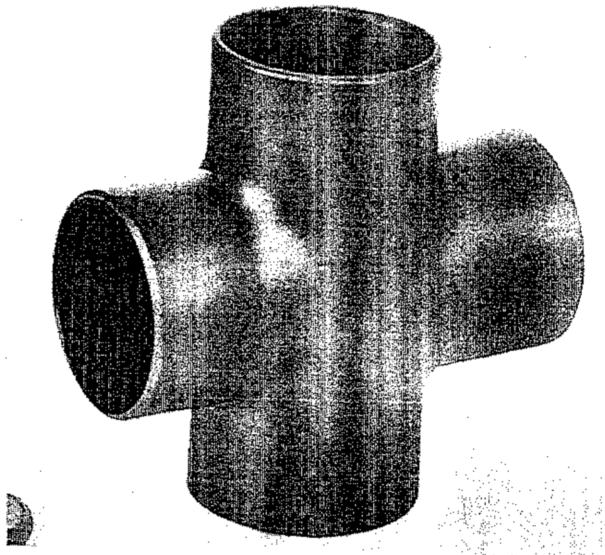


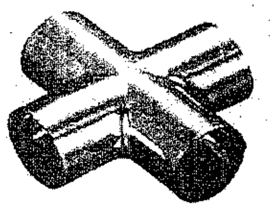
N/B Equal/straight Tee and Reducing Tee (see above)

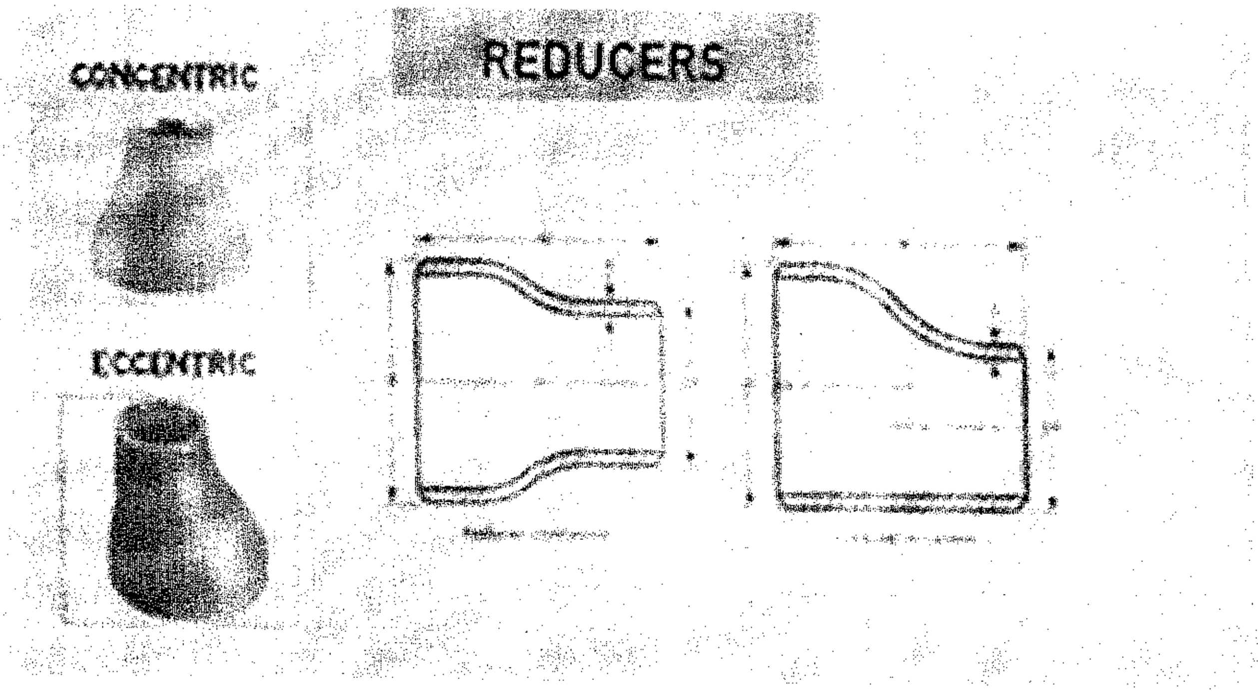
CROSS

Socket Weld Cross





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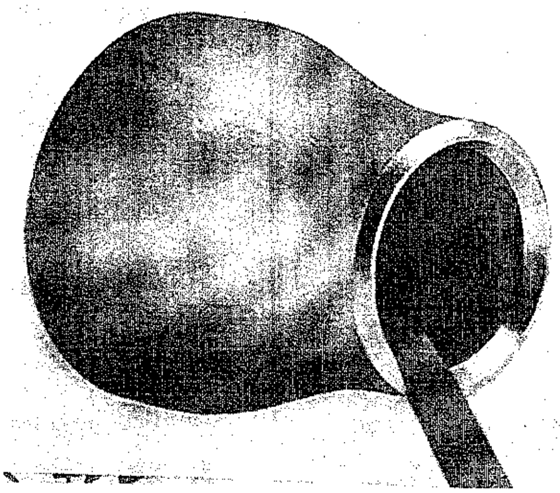
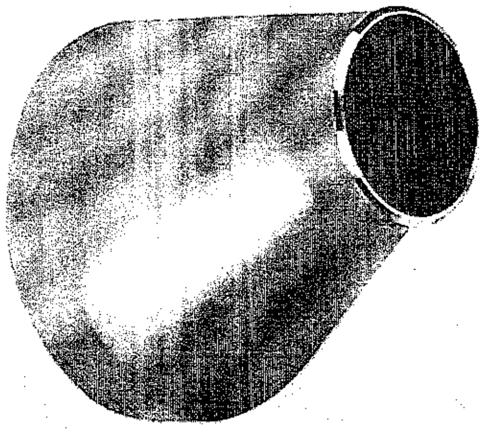
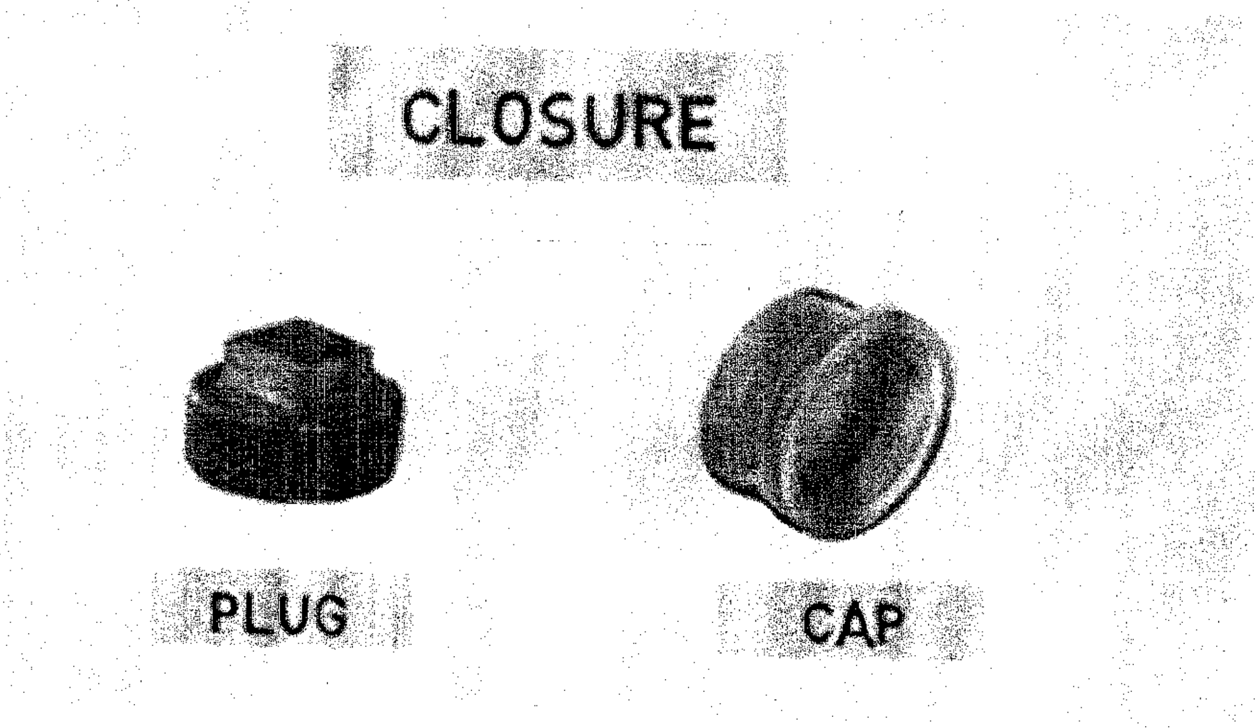
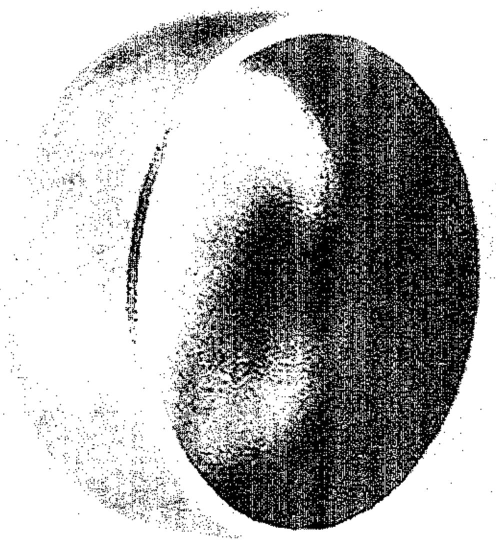
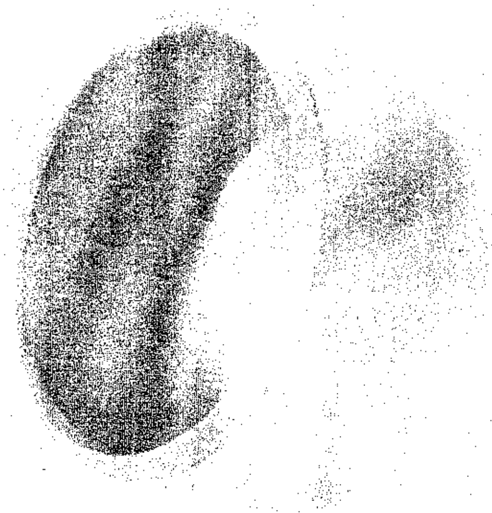


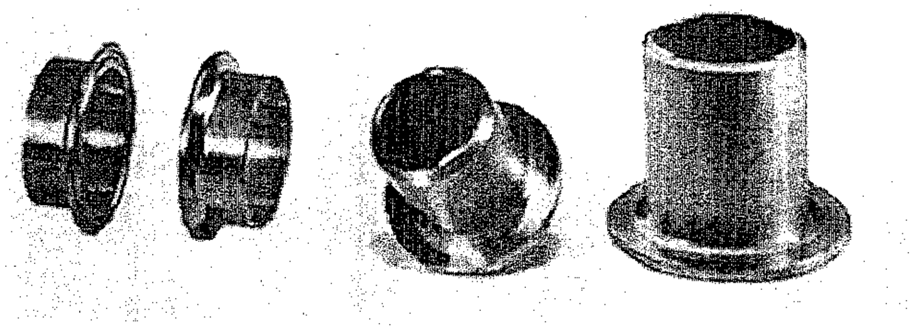




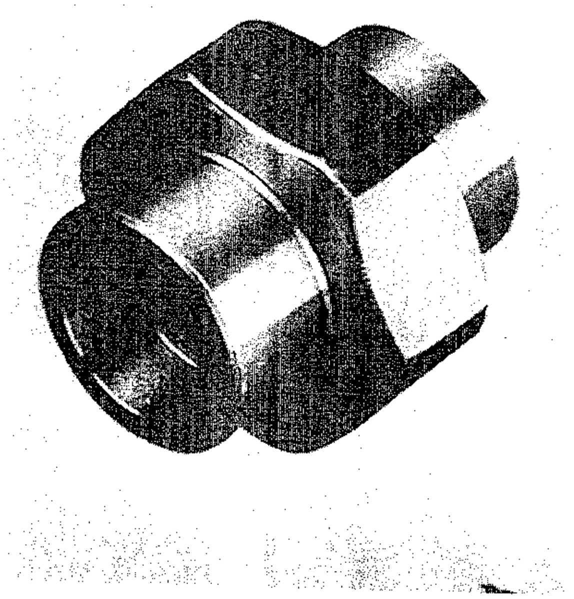


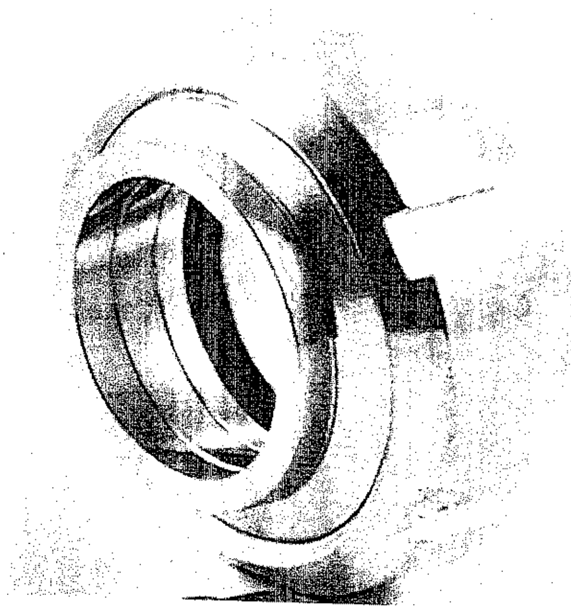


Flanges (Long and Short flanges)

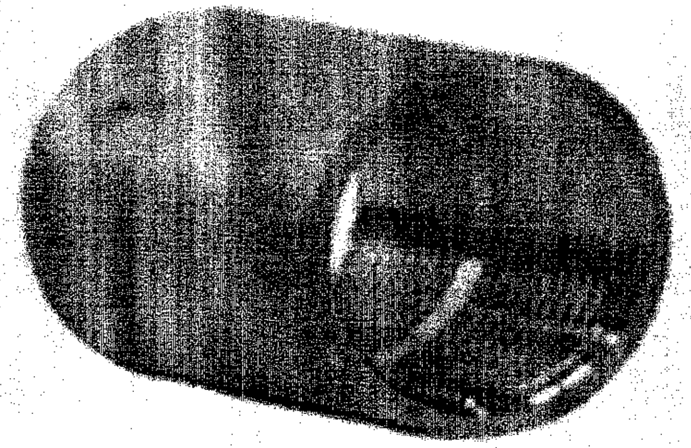


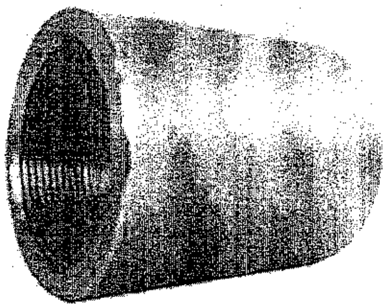
Union

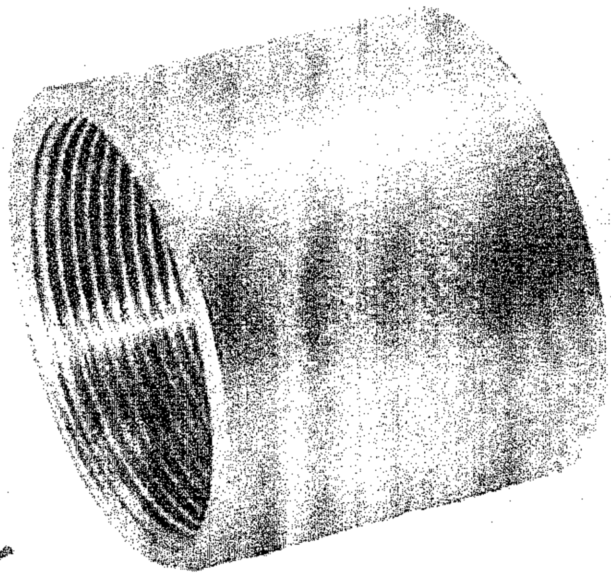




COUPLINGS

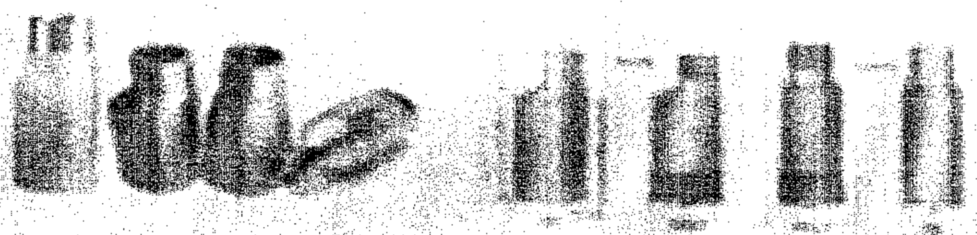






1) FULL Coupling
2) Half "
3) Reducing coupling

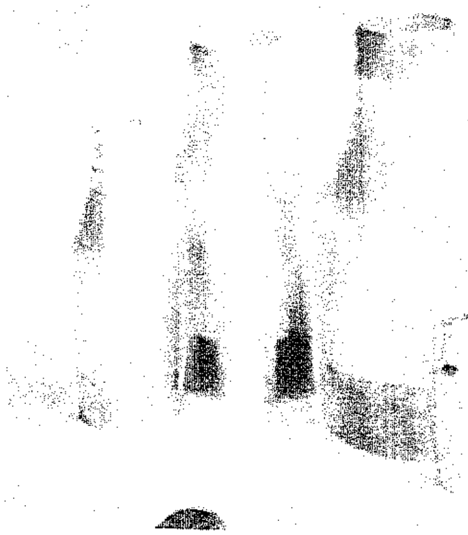
SWAGE NIPPLE



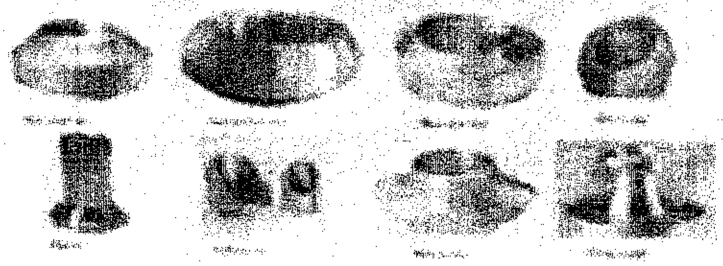
THE DIFFERENT END CONNECTIONS AVAILABLE ARE

→ PBE - PLAIN BOTH ENDS
→ PLE - PLAIN LARGE ENDS
→ PSE - PLAIN SMALL END
→ BLE - BEVELED LARGE END

y (k)



OLETS

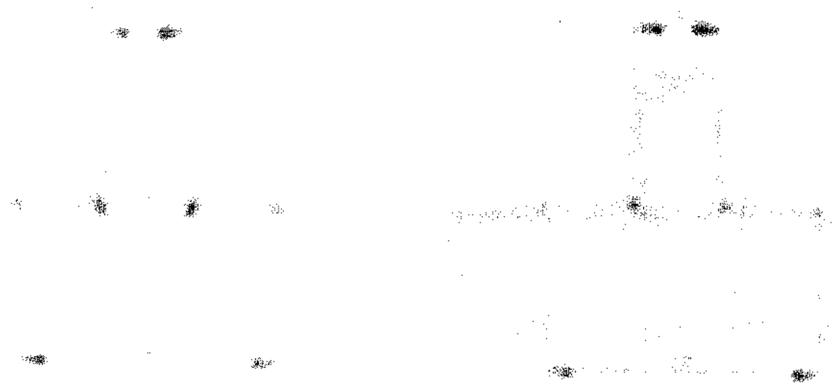


SOME IMPORTANT TYPES OF OLETS ARE

- → BUTT-WELDOLET
- → SOCKET WELDOLET
- → THREADED OLET
- → ELBOW OLET
- → NIPPLE OLET
- → LATERAL OLET
- → SWEEPOLET
- → FLANGE OLET

Play (k)

STUB-IN & STUB ON



STUB-IN

STUB-ON

These fittings, along with others, provide the necessary versatility, adaptability, and reliability in plumbing and piping systems, allowing for efficient and effective fluid transport and control. The specific choice of fittings depends on the system requirements, pipe dimensions, fluid characteristics, and industry standards.

Codes and Standards used in Pipe Fittings

Codes and **standards** play a crucial role in ensuring the quality, safety, and compatibility of pipe fittings used in various industries and applications. Here are some of the key codes and standards commonly used in the realm of pipe fittings:

1. **ASTM Standards**: The American Society for Testing and Materials (ASTM) develops and publishes standards for various materials, including metals, plastics, and rubber used in pipe fittings. These standards specify the material properties, dimensions, testing methods, and performance requirements for pipe fittings.
2. **ASME B16 Standards**: The American Society of Mechanical Engineers (ASME) publishes the B16 series of standards, which cover different types of pipe fittings. The most commonly referenced standards from this series include ASME B16.9 (Factory-Made Wrought Steel Butt-Welding Fittings), ASME B16.11 (Forged Fittings, Socket-Welding and Threaded), and ASME B16.5 (Pipe Flanges and Flanged Fittings).
3. **MSS Standards:** The Manufacturers Standardization Society of the Valve and Fittings Industry (MSS) publishes standards related to pipe fittings. The MSS standards cover various aspects, including materials, dimensions, testing, and marking requirements for specific types of fittings, such as MSS SP-75 for high-strength, wrought, butt-welding fittings and MSS SP-97 for integrally reinforced forged branch outlet fittings.
4. **ANSI/ASME Standards**: The American National Standards Institute (ANSI) in conjunction with ASME establishes standards for pipe fittings. ANSI/ASME B31.1 (Power Piping) and ANSI/ASME B31.3 (Process Piping) provide guidelines for the design, fabrication, and installation of piping systems, including the use of pipe fittings.
5. **ISO Standards:** The International Organization for Standardization (ISO) develops international standards for various products, including pipe fittings. ISO standards provide specifications for

materials, dimensions, and performance characteristics of pipe fittings, ensuring global compatibility and interchangeability.

6. **API Standards**: The American Petroleum Institute (API) publishes standards specific to the oil and gas industry, including standards for pipe fittings used in pipelines and refineries. Examples of API standards related to pipe fittings include API 5L (Line Pipe), API 6D (Pipeline Valves), and API 600 (Steel Gate Valves).

These are just a few examples of the many codes and standards used in pipe fittings. It's important to consult the specific code or standard applicable to your industry, application, and geographical region to ensure compliance and proper selection of pipe fittings for your specific needs.